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AI orchestrator: How recommendation algorithms shape complementor strategy and market equality

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Abstract

Research Summary: This study investigates how AI recommendation algorithms shape complementor strategies and market equality on digital platforms. Using two quasi-natural experiments from a food-sharing platform, we examine impacts of sequential algorithmic upgrades: from a location-based baseline to popularity-based (PopRec) and then personalization-based (PersRec). Analyses of over 1.7 million observations reveal that PopRec drives complementors to concentrate on a few offerings, while PersRec encourages new product introduction; yet these strategic shifts come at the expense of one another. Furthermore, PopRec reduces revenues of superstars but boosts revenues for long-tail complementors, enhancing market equality. Conversely, PersRec exacerbates market inequality by asymmetrically benefiting superstars. By bridging platform orchestration and AI frontiers, this study demonstrates the strategic potential of AI in platform management while underscoring the importance of algorithmic accountability.

Managerial Summary: Digital platforms increasingly deploy AI recommendation algorithms to manage ecosystem performance. This study reveals that these algorithms also function as effective orchestration mechanisms, incentivizing and shaping complementor offering strategies at scale. We find that popularity-

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based algorithms incentivize complementors to specialize and concentrate on a few core product offerings, while personalization-based algorithms foster broader new product introduction. Our results demonstrate the vital strategic value of AI for managing platform ecosystems. Crucially, both algorithms present inherent trade-offs for complementor strategies and trigger unforeseen market dynamics. Platform owners must meticulously design and implement algorithmic systems, balancing ecosystem generativity with operational control and considering far-reaching market inequality implications.

KEYWORDS

artificial intelligence, digital platform, market equality, platform orchestration, recommendation algorithm

1 | INTRODUCTION

Digital platforms rely on platform orchestration: the deliberate design of incentives and rules to align autonomous complementors with platform-level goals (Rietveld et al., 2021; Rietveld & Schilling, 2021). Effective orchestration is critical to sustaining innovation, coordination, and value creation across diverse platform ecosystems such as sharing economy platforms, gig marketplaces, e-commerce sites, and app stores (Chen, Tong, et al., 2022; Kretschmer et al., 2022; Leong et al., 2024). However, most forms of platform orchestration are shaped by a fundamental tension between *generativity*[1] and *control* (Staub et al., 2022; Wareham et al., 2014; Yoo et al., 2012). Generativity reflects a platform's capacity to foster distributed innovation, diversity, and vitality through autonomous complementors, whereas control ensures quality, coherence, and stability (Boudreau, 2010, 2012; Cennamo & Santaló, 2019). While this tension is widely recognized, less is known about how platform owners can design orchestration mechanisms that effectively balance these opposing forces (Rietveld & Schilling, 2021). Because a platform's generativity depends on the voluntary and heterogenous contributions of complementors, managing this balance ultimately requires orchestrating their micro-level offering strategies (Koo & Eesley, 2021; Wareham et al., 2014). Yet, complementors differ markedly in motivations, capabilities, and goals, making it increasingly difficult to develop orchestration mechanisms that are both adaptive and scalable (Boudreau & Jeppesen, 2015; Zhang et al., 2022).

Recent advances in artificial intelligence (AI) offer new insights into the generativity-control tension inherent in platform orchestration. In practice, AI algorithms are increasingly leveraged to manage complementors by calibrating incentives, coordinating exchanges, and curating boundary resources (Apple, 2025; TikTok, 2025; Uber, 2022). With the ability to learn from

[1]Throughout this study, our use of "generativity" is distinct from recently popular notion of "generativity" in the context of generative AI, which denotes the ability of AI technologies themselves to create and generate novel, high-quality content.

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large-scale behavioral data and autonomously adjust decision rules (Hillebrand et al., 2025; Krakowski et al., 2023), AI systems function not merely as technical optimizers but as orchestration mechanisms that dynamically influence complementor behavior and market dynamics at scale (Cram et al., 2022; Mayer et al., 2025). Among these systems, AI recommendation algorithms form a particularly important orchestration lever for platform owners to navigate the dynamics between generativity and control. Complementors respond strategically because visibility granted through recommendation is a valuable reward, motivating them to adjust their offering strategies to compete for algorithmic exposure (Qian & Jain, 2024; Zhou & Zou, 2023). While extensive research has examined how recommender systems shape consumer demand (Adomavicius et al., 2018; Li et al., 2022; Liu & Cong, 2023), far less is known about how different AI recommendation designs function as orchestration mechanisms that guide complementor strategies and redistribute value across the marketplace. Moreover, algorithmic changes may propagate across interconnected complementors, generating divergent outcomes for superstars and long-tail complementors and thereby market inequality concerns (Malgonde et al., 2020; O'Connor, 2025; Zhang et al., 2021).

In our study, we examine two core AI recommendation algorithm designs that represent the typical evolutionary stages on digital platforms: a popularity-based recommendation algorithm (PopRec), which ranks complementors based on aggregated customer ratings, and a personalization-based recommendation algorithm (PersRec), which tailors exposure based on individual user preferences (Adomavicius et al., 2022; Sun et al., 2024). Our empirical setting, a major food-sharing platform, successively implemented two algorithmic upgrades. PopRec resembles traditional rule-based, non-personalized approaches, whereas PersRec employs machine learning techniques to infer consumer preferences from behavioral data such as browsing history and purchase patterns. Because these two algorithms allocate visibility through distinct generativity-control logics and thus create different competitive incentives for complementors, strategically designing and upgrading them enables platform owners to steer complementor behaviors through "digital orienting" (Leong et al., 2024). Importantly, both algorithms exemplify predictive AI, which learns from historical data to make domain-specific decisions (Choudhury et al., 2020; Jia et al., 2024; Luo et al., 2024), distinguishing them from emerging generative AI (Doshi et al., 2025; Raisch & Fomina, 2025).

More specifically, our study investigates: (1) *how different recommendation algorithms (PopRec and PersRec) influence complementors' product offering strategies, particularly the trade-off between focusing on core offerings and introducing new ones*; and (2) *how these algorithmic upgrades reshape market equality by redistributing business outcomes among superstars and long-tail complementors*. These questions directly speak to how predictive AI, by encoding distinct decision logics, shapes strategic adaptation and ecosystem evolution (Hillebrand et al., 2025; Rietveld et al., 2020).

We draw on a proprietary dataset from a major food-sharing digital platform that captures the full population of complementors. This comprehensive dataset enables us to analyze the frequent adjustments in offering strategies by individual complementors, as well as the overall distributional equality of market outcomes. We construct panel datasets at the complementor-day level, comprising 1,670,029 observations for the PopRec algorithm upgrade and 1,759,102 observations for the PersRec algorithm upgrade. By leveraging these platform-wide, exogenous AI algorithm changes, we employ regression discontinuity in time (RDiT) designs to identify the impacts of the algorithmic upgrades on complementors. We find that complementors are incentivized by PopRec to specialize and concentrate on a few core offerings, while PersRec encourages them to pursue broader new product introduction. These strategic shifts come at the

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expense of one another and exhibit heterogeneity among complementors under different competitive conditions. Further, we find that PopRec reduced the revenue of prior superstars while boosting revenues for long-tail complementors, thus enhancing market equality (as measured by Lorenz curve and Gini index). By contrast, PersRec exacerbates market inequality by benefiting superstars more so than the long tail.

This study makes several important contributions. First, it addresses a critical gap in understanding how AI recommendation algorithms function as orchestration tools to navigate the interplay between platform generativity and control (Chen, Tong, et al., 2022; Rietveld & Schilling, 2021; Staub et al., 2022). We show that AI recommendation algorithms can be employed to nudge complementors' strategies toward platform-level goals, thereby facilitating platform orchestration. Such orchestration relies on complementors' autonomous, strategic decision-making rather than contractual or hierarchical control, and thus entails an inherent strategic tradeoff. Second, it underscores the value of AI's technological advantages in platform ecosystem management. We elucidate the substantial strategic potential of AI by demonstrating how recommendation algorithms enable automatic, data-driven promotion of complementors and facilitate the large-scale calibration of complementor strategies. In this sense, our work contributes to the recently burgeoning literature stream on strategic AI (Doshi et al., 2025; Krakowski et al., 2023; Tong et al., 2021). Third, our findings contribute to the emerging literature on the market equality and policy implications of AI algorithms (Koo & Eesley, 2021; Rahman et al., 2024; Zhang et al., 2021). We enrich this research by illustrating how AI recommendation algorithms unevenly impact complementors' strategies and performance and reshape the competitive landscape. Our findings highlight the organizational complexities and value capture asymmetries that can arise from AI-driven orchestration (Rietveld et al., 2020), underscoring the need for algorithm accountability and guiding policymakers to ensure market equality of AI-based platform orchestration (Furman & Seamans, 2019; Grote et al., 2024).

2 | THEORY AND HYPOTHESES

2.1 | Platform orchestration tension

Platform owners play a crucial role in orchestrating autonomous complementors and ensuring the functioning and sustainability of platform ecosystems (Chen, Tong, et al., 2022; Jacobides et al., 2018; Kretschmer et al., 2022). Unlike traditional governance practices based on direct control and enforcement (He et al., 2025; Zhang et al., 2022), platform orchestration relies on incentive-based mechanisms that steer complementors toward desired behaviors while preserving their autonomy (Gastaldi et al., 2024). Studies have identified diverse orchestration mechanisms, such as competing directly with complementors (Kapacinskaite & Mostajabi, 2024; Wen & Zhu, 2019), providing boundary resources (Mayer et al., 2025), and granting selective endorsements (Rietveld et al., 2019), certifications (Rietveld et al., 2021), awards (Foerderer et al., 2021), and venture capital (Van Angeren & Karunakaran, 2023). These mechanisms act as proxies for the platform's influence by shaping which complementary contributions are encouraged and rewarded, and thus are pivotal for managing both the ecosystem's value creation and the platform owner's own value appropriation (Hunt et al., 2025; Rietveld et al., 2020).

A persistent challenge in platform orchestration lies in balancing between *generativity* and *control* (Staub et al., 2022; Wareham et al., 2014; Yoo et al., 2012). Platform owners must foster

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bottom-up innovation and creativity while maintaining coherence, quality, and efficiency (Boudreau, 2010, 2012; Engert et al., 2025). Excessive generativity risks a proliferation of low-quality offerings, a phenomenon reminiscent of the notorious "Atari shock" (Kretschmer et al., 2022), whereas excessive control can stifle innovation. This inherent tension makes it difficult to orchestrate complementor strategies effectively. Efforts to emphasize one dimension often disrupt the balance, interfering with complementors' strategic configurations and triggering unintended ecosystem-wide consequences (Chen, Yi, et al., 2022; Kapacinskaite & Mostajabi, 2024). As Rietveld and Schilling (2021) note, the generativity-control tension has been underexplored in existing platform research, limiting our understanding of the boundary conditions and outcomes of orchestration mechanisms.

This tension is further complicated by complementor heterogeneity. Complementors differ in motivations, objectives, and capabilities (Boudreau & Jeppesen, 2015; McIntyre & Srinivasan, 2017), which causes orchestration mechanisms to affect them unevenly. These disparities can propagate across the market, advantaging some complementors while disadvantaging others and raising concerns about economic inequality. For example, Koo and Eesley (2021) show that rural sellers, due to limited information access, are less able than urban sellers to adapt to major design changes, unintentionally widening the digital divide. Such value redistribution may render some seemingly effective orchestration mechanisms counterproductive. In this study, we seek to examine whether and how AI recommendation algorithms can help platforms navigate the generativity-control tension. Furthermore, drawing upon research on complementor heterogeneity and market inequality (Polidoro & Yang, 2024; Rahman et al., 2024; Skiti et al., 2022), we also explore the downstream market equality dynamics that such AI-empowered platform orchestration could engender.

2.2 | AI and recommendation algorithm on digital platforms

AI technologies have attracted growing attention from strategy and management scholars for their transformative potential in organizational decision-making and value creation. AI not only complements human intelligence by enhancing analytical capabilities and productivity (Berg et al., 2023; Choudhury et al., 2020; Jia et al., 2024) but also enables firms to address previously intractable strategic challenges (Doshi et al., 2025; Luo et al., 2024; Raisch & Fomina, 2025), thereby creating new sources of competitive advantage (Krakowski et al., 2023). Yet, its proliferation also raises concerns about job displacement, security vulnerabilities, and ethical accountability (Furman & Seamans, 2019; Grote et al., 2024; Tong et al., 2021).

The core benefit of AI lies in its capability to process vast amounts of data to automate tasks, detect patterns, and make predictive decisions at scale (Hillebrand et al., 2025; Krakowski et al., 2023). These capabilities are consequential on digital platforms, where standardized infrastructure and continuous data generation create fertile conditions for AI-driven orchestration (Chen, Tong, et al., 2022; Sun et al., 2024). Leveraging AI systems, platform owners can coordinate and influence the behaviors of multi-sided participants in an automated, data-driven, and scalable manner, thereby enhancing both managerial efficiency and user experience (He et al., 2025; McDaid et al., 2023; Zhang et al., 2021). Particularly, AI technologies can tailor and deliver behavioral nudges (Cram et al., 2022; von Zahn et al., 2025) that shape action possibilities while preserving participants' autonomy (Leong et al., 2024).

AI recommendation algorithms are among the most ubiquitous AI applications on digital platforms. By effectively reducing consumer search costs and mitigating information

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asymmetry, these algorithms are widely recognized for enhancing consumer experience (see Supporting Information Appendix A for a comprehensive review). Importantly, algorithmic recommendations also play a managerial role by enabling automated sales promotion, which makes them a potentially powerful tool for platforms to orchestrate the visibility and supply of complementors. Digital platforms such as Airbnb, Amazon, eBay, Forbes, Netflix, and YouTube have typically evolved through two critical stages of AI recommendation algorithms (Supporting Information Appendix B). The first, the PopRec algorithm, ranks offerings according to aggregated user ratings, rewarding complementors that align with widely approved preferences (Adomavicius et al., 2022). The second, the PersRec algorithm, employs machine learning to infer individual users' preferences from behavioral data such as browsing and purchase histories (Li et al., 2022; Wan et al., 2024). Whereas PopRec reinforces established popularity patterns and promotes conformity with dominant demand signals, PersRec encourages experimentation and differentiation by catering to niche preferences. These contrasting logics embody distinct promotional goals of platform owners, and thus, play varying roles in platform orchestration.

2.3 | Platform orchestration through AI algorithmic recommendations

At their core, AI recommendation algorithms define the rules of visibility allocation, determining which complementors, based on their performance or characteristics, are prioritized and thus granted greater exposure on the platform. Because visibility is a scarce and highly valued resource that facilitates transactions (Oestreicher-Singer & Sundararajan, 2012), it effectively functions as a form of algorithmic reward. To obtain this reward, complementors strategically adjust their offerings to align with algorithmic preferences, actively "competing for the algorithmic recommendation" (Li et al., 2020; Qian & Jain, 2024; Zhou & Zou, 2023). The automatic and adaptive feedback loop of these algorithms allows rankings and reward structures to evolve continuously, ensuring that complementor responses yield immediate and measurable consequences.

Building on such organic strategy adaptations, the deliberate design of recommendation algorithms can be employed by platform owners to strategically nudge complementors' behaviors. In this sense, AI recommender systems act as powerful instruments of algorithmic orchestration, a form of behavioral steering also referred to as "digital orienting" (Leong et al., 2024). By tailoring the incentive mechanisms embedded within recommendation algorithms to align with platform orchestration objectives (Kretschmer et al., 2022; Malgonde et al., 2020), platforms can subtly shape complementors' competitive strategies through "soft control" (Tiwana, 2013; Tiwana et al., 2010). This approach extends Rietveld et al.'s (2019) concept of selective promotion from direct managerial intervention to algorithmic decision-making, which is automated yet strategically aligned with the platform's generativity-control objectives.

2.4 | Hypotheses on the impacts of AI recommendation algorithms on complementors' offering strategies

In this section, we develop hypotheses on how the two most prevalent implementation phases of AI recommendation algorithms (i.e., PopRec and PersRec) influence complementors' offering

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strategies. First, the PopRec algorithm, which amplifies popularity signals by prioritizing offerings with higher customer ratings, incentivizes complementors to enhance their visibility by optimizing factors that drive positive customer feedback (Adomavicius et al., 2022). Given that consumer ratings reflect aggregated evaluations of quality, reliability, and customer satisfaction (Li, 2018), complementors are encouraged to strengthen their core offerings to secure positive evaluations within a controllable scope (Voss et al., 2008). In doing so, they channel resources toward refining established product competencies and reinforcing their market positions; thereby complementors are more likely to specialize and concentrate on a few core offerings in response to the PopRec algorithm.

However, this same mechanism can dampen generativity. New product introductions often entail uncertainty and ambiguous valuations (Boudreau, 2012; Ploog & Rietveld, 2025; Rietveld & Eggers, 2018), and the PopRec algorithm's visibility logic exacerbates the associated risk. Because newly launched products lack prior ratings and market feedback, adding them to the product portfolio not only yields no additional algorithmic visibility but may also undermine the market reputation established by existing products, decreasing the recommendation potential for complementors. This implies that complementors who diversify their portfolios may incur potential algorithmic penalties, rendering the experimentation with and introduction of new offerings exceptionally costly. Consequently, while complementors are incentivized to fortify their market-tested core product competencies, they are less inclined to divert resources toward riskier new product introductions (i.e., less likely to introduce new products) in response to the PopRec algorithm. Thus:

> **Hypothesis (H1).** Under PopRec (as compared to no recommendation), complementors are (a) more likely to specialize and concentrate on a few core offerings and (b) less likely to introduce new products.

In contrast, the PersRec algorithm prioritizes personalized relevance by matching consumers with complementors that best fit their individual needs and tastes (Adomavicius et al., 2022; Wan et al., 2024). The contrasting algorithmic logic can lead PersRec to produce strategic effects that are opposite to those of PopRec, which precedes PersRec in our setting and across the algorithmic lifecycles of many digital platforms. Because visibility is granted to complementors capable of catering to a broader spectrum of consumer preferences, the PersRec algorithm incentivizes them to expand their product portfolios by introducing new offerings that appeal to diverse and fragmented demand (Klingebiel & Rammer, 2014). By expanding their assortments, complementors can reach additional consumer segments and increase transaction opportunities (Kekre & Srinivasan, 1990; Meyer et al., 2024), particularly in markets characterized by high preference heterogeneity (Ploog & Rietveld, 2025; Rietveld & Eggers, 2018). This visibility logic thus promotes generativity by encouraging experimentation and the introduction of new complements. In other words, complementors are more likely to engage in broader new product introduction in response to the PersRec algorithm.

However, under the PersRec algorithm, as complementors extend their product portfolios, they often face inherent trade-offs. Specifically, developing new offerings requires reallocating limited financial, operational, and managerial resources (Tae et al., 2020; Voss et al., 2008; Wen & Zhu, 2019). Given that their resource base cannot be expanded immediately, complementors may scale back investment in core offerings to spread it across broader market segments, thereby less likely focusing on a few core product offerings (Boudreau, 2012; Voss et al., 2008) in response to the PersRec algorithm. Therefore, we hypothesize:

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Hypothesis (H2). Under PersRec (as compared to PopRec), complementors are (a) more likely to engage in broader new product introductions and (b) less likely to focus on a few core offerings.

2.5 | Hypotheses on the impacts of AI recommendation algorithms on market equality

Although the complementors' strategic responses make the algorithmic orchestration feasible, the AI algorithms can expose different complementors, such as superstars and long-tail complementors, to heterogeneous shocks (Brynjolfsson et al., 2010; Lee & Hosanagar, 2019). Such exposure may elicit varied responses and reshape the existing competitive landscape, whereby the strategic responses predicted in H1 and H2 are likely to act as mechanisms that have downstream impacts on market equality of the platform.

More specifically, superstar complementors who already capture organic traffic and broad consumer recognition, receive less additional visibility under the PopRec algorithm that prioritizes offerings with higher customer ratings (Benner & Waldfogel, 2023; Brynjolfsson et al., 2010). Also, the competitive threat posed by PopRec could further reinforce superstars' strategic conservatism, making them less inclined to pursue horizontal growth by broadening offerings in ways that increase uncertainty (Lavie et al., 2010). Thus, their *marginal* returns from the rating-driven PopRec recommendations are lower. In contrast, PopRec algorithm also enhances the visibility of previously underexposed yet well-rated complementors. This mechanism disproportionately benefits prior long-tail complementors (relative to superstars) who, despite offering quality products/services, often struggle with salience and recognizability (Brynjolfsson et al., 2006, 2010). Also, long-tail complementors, who typically serve niche markets, are better positioned to specialize and concentrate on their core product competencies in response to the algorithm (Brynjolfsson et al., 2006; Sorenson et al., 2006). Thus, these long-tail complementors can generate more incremental consumer attention and participation, and their *marginal* returns from PopRec recommendations are higher. In summary, PopRec likely shifts performance gains toward previously disadvantaged long-tail complementors, narrowing performance disparities across the complementor population and increasing market equality on the platform. Hence, we hypothesize:

Hypothesis (H3). PopRec (as compared to no recommendation) will increase market equality on the platform.

In stark contrast, as the PersRec algorithm allocates visibility based on personalized relevance, matching users with complementors whose offerings best align with individual preferences (Adomavicius et al., 2018; Li et al., 2022; Wan et al., 2024), both prior superstars and long-tail complementors may benefit from the PersRec algorithm. However, inherent differences in endowments, capabilities, and experience make prior superstars more likely to capture these benefits. That is, with richer historical transaction data, market-validated product portfolios, and more adaptive operational capabilities, superstars are more capable of aligning with individual demand profiles and responding to algorithmic cues (Meyer et al., 2024). Their accumulated experience allows them to leverage PersRec's feedback loops more efficiently by introducing tailored new products and adjusting offerings with greater agility (Choi & Shepherd, 2004; Lavie et al., 2010). Consequently, PersRec reinforces the advantages of

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superstars while widening the performance gap between them and their long-tail counterparts. Indeed, some demand-side focused research documented that, while PersRec has indeed fostered niche sales, revenue in many online marketplaces remains concentrated among a few superstar complements (Lee & Hosanagar, 2019; Tan et al., 2017). Thus, overall, we expect PersRec to amplify the advantages of superstars, potentially enlarging the performance disparities across the complementor population and worsening market inequality on the platform.

Hypothesis (H4). PersRec (as compared to PopRec) will exacerbate market inequality on the platform.

3 | RESEARCH CONTEXT AND DATA

3.1 | Research context

We acquire a rich proprietary dataset from a major food-sharing digital platform in China. The app-based platform was launched in October 2014 and is the largest home-cooked food-sharing platform in the local market, with over 1 million active complementors and 10 million active consumers each month. This digital platform matches complementors who cook dishes in their home kitchens with nearby demand-side consumers who order meals. Unlike UberEats and Grubhub, whose complementors are professional restaurants, this platform hosts amateurs and small-scale entrepreneurs. These individuals exploit under-utilized kitchen resources and home cooking skills for business exchanges. In Supporting Information Appendix B, we provide a comprehensive overview of the platform, and we compare it with other typical digital platforms.

The food-sharing platform implemented two algorithm upgrades (a PopRec algorithm on July 1, 2016, and a PersRec algorithm on October 13, 2016) on top of the baseline geo-fencing interface (no recommendation). Since its inception, the platform has used geo-fencing mobile technology, which facilitates matching complementors and consumers based on their location proximity. This choice was primarily due to the platform's lack of data at the nascent stage, which hampered the implementation of genuine recommendation algorithms (a cold start problem). As the platform collected more demand-side data on customers' review ratings over time, it decided to launch a PopRec algorithm. The PopRec algorithm employs a non-personalized, product-rating-based ranking method to generate recommendation lists. The implementation of the PopRec algorithm on top of the geo-fencing interface resulted in the highest-rated nearby complementors being displayed on the app's main screen by default.[2] As a PopRec algorithm recommends complementors based on consumer review ratings or the "wisdom of the crowd," its implementation was intended to improve customer satisfaction by prioritizing top-rated complementors near consumers.

Three months later, to further enhance the search experience of consumers, the platform implemented a PersRec algorithm based on individual consumers' past browsing, searching, and purchasing behaviors on the app.[3] Specifically, this algorithm used personal data on each

[2]At this stage, the app's homepage defaulted to displaying recommendation results based on the PopRec algorithm. Additionally, consumers had the option to revert to the previous geo-fencing interface if they preferred.

[3]The sequential rollout from PopRec to PersRec is a common practice among platforms, particularly during early-stage growth. The primary reasons for this include cold start and data availability, technical and computational considerations, and operational effectiveness and platform scale.

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consumer's food search behavior, cuisine preferences, dish price, time of order, and purchase recency, frequency, and quantity. After extensively testing and adjusting its IT systems, the app platform confirmed that the PersRec algorithm worked and was able to recommend personalized dishes based on each individual's personal preferences. The PersRec algorithm was launched on top of the PopRec and geo-fencing interfaces to ensure that the best-suited nearby complementors who can satisfy an individual user's personal preferences are displayed on the app's main screen by default.[4] Supporting Information Appendix B provides examples of the three algorithms to aid understanding and generalizability of our context to other recommendation algorithms, digital platforms, and system design.

As PopRec and PersRec algorithms were exclusively engineered and launched by the platform, there was no pre-announcement. The platform manager confirmed that it was impossible for complementors or consumers to know *ex-ante* about the new algorithms and to have preemptively adopted strategic behaviors that might confound our estimates of the algorithms' impacts. In this sense, the algorithm upgrades were *exogenous* shocks to both complementors and consumers. Once implemented, the nature of the algorithms was prominently displayed at the top of landing pages, making them easily identifiable by both consumers and complementors. The platform's executives also confirmed that there were no other major functional changes, interface design changes, or marketing promotions on the platform during the launch of these two algorithms. Furthermore, the platform manager noted that the timing of the algorithm rollout was not a deliberate consideration, owing to the inherent difficulty in predicting the potential reactions of heterogeneous complementors. This provides us with a credible quasi-natural experimental setting to accurately identify the net effects of the platform's algorithm upgrades. Notably, following each algorithm's implementation, customers had the option to proactively revert their software configurations to the previous algorithms. Should complementors anticipate such consumer behavior, they are likely to be more conservative in their strategic adjustments. This conservatism, in turn, suggests that our estimates of algorithmic impacts may be lower bounds relative to a setting where the algorithm is uniformly enforced.[5]

3.2 | Data and variables

The extensive dataset provided by the food-sharing platform consists of all complementors in a major city in China over a period of 9 months. Our observation window encompasses three implementation stages of the recommendation algorithms: *No recommendation* (91 days), *PopRec* (104 days), and *PersRec* (80 days), which are used to construct two panels: Panel A employs the No recommendation and PopRec stages to investigate the impact of PopRec upgrade as an exogenous shock; Panel B employs the PopRec and PersRec stages to investigate the impact of PersRec upgrade as an exogenous shock. These two panels are constructed at the complementor-day level, with observations aggregated from more granular order-level, dish-level, and review-level data. The daily panel data includes 1,670,029 observations for the PopRec upgrade and 1,759,102 observations for the PersRec upgrade. Figure 1 summarizes the timeline of the algorithm upgrades and our data collection windows.

[4]At this stage, the app's homepage defaulted to displaying recommendation results based on the PersRec algorithm. Additionally, consumers had the option to revert to the previous PopRec algorithm or geo-fencing interface if they preferred.

[5]This partial exposure environment constitutes a meaningful boundary condition. Indeed, many digital platforms (e.g., Netflix, Amazon, YouTube) often employ fully automated "black box" systems, where users cannot revert to prior algorithms. Such irreversibility may provoke more aggressive strategic responses from complementors.

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3.2.1 | Independent variables

We construct the binary variables PopRec_t and PersRec_t to denote whether the time point t of each observation falls in the pre-implementation (assigned a value of 0) or post-implementation (assigned a value of 1) periods of the PopRec and PersRec algorithms, respectively.

3.2.2 | Dependent variables

We delineate two key dimensions of complementors' offering strategies (Voss & Voss, 2013). First, we focus on the concentration of offerings within a particular category. This reflects the extent to which complementors dedicate efforts to refine, specialize, and optimize in a focused area, often involving capitalization of their established core product competencies (Voss et al., 2008). We measure this concentration using the Herfindahl–Hirschman Index (HHI) of dish genres, calculated at the complementor-day level by squaring the share of each dish genre within a complementor's daily offerings and summing the squared shares (Koo & Eesley, 2021; Kumar, 2009). To capture the changes in the concentration of dish offerings over time, we perform a first-order difference on the HHI (denoted by ΔHHI), which serves as our main dependent variable. For robustness checks, we calculate HHI based on meat ingredients, cooking methods, and customer-recommended dishes. We also utilize the breadth of the product portfolio dimensions as alternative measures.

Second, we focus on complementors' new product introductions. This reflects the extent to which complementors explore, experiment with, and expand their product offerings beyond existing product portfolios to create new business opportunities (Voss & Voss, 2013). Our primary metric is the daily number of newly introduced dishes. To enable granular analysis and robustness verification, we also calculate the daily numbers of newly adopted meat ingredients and newly adopted cooking methods.

3.2.3 | Control variables

We construct a comprehensive set of control variables at various levels to rule out alternative explanations. Specifically, we control for menu characteristics (number of dishes, average price), review features (number of reviews, prior review ratings), coupon features (value of coupon offered), kitchen features (gender, age, tags, delivery radius and fees), market competition,

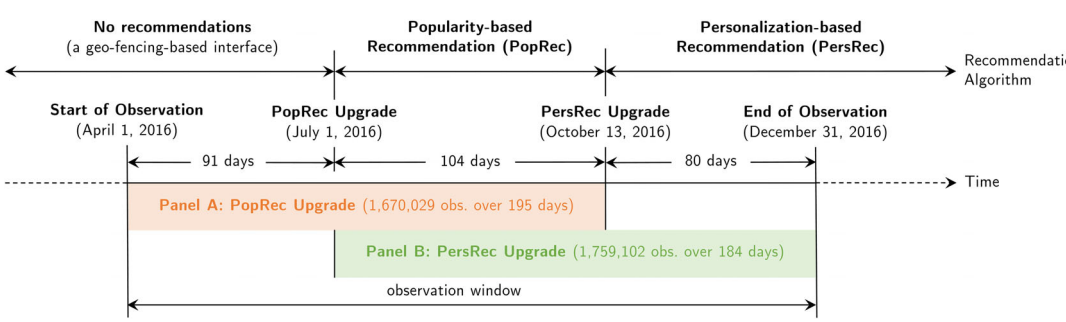

FIGURE 1 Timeline of the algorithm upgrades and data collection windows.

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consumer attributes (disclosure of gender/age), platform advertising, etc. The detailed description of the control variables is presented in Table 1.

3.2.4 | Data descriptives

Table 1 provides a descriptive summary of the variables. On average, each complementor receives CNY 47.663 and CNY 42.278 in daily revenue in Panels A and B, respectively. More than 70% of the complementors on the platform are female, and the average age of the complementors is 41.548. These statistics align with the nature of the food-sharing platform, in which many of the cooks are middle-aged housewives who spend their spare time preparing and serving home-cooked dishes for extra income.

4 | ANALYSIS AND RESULTS

4.1 | Results for the impacts of AI recommendation algorithms on complementors' offering strategies

4.1.1 | Identification strategy: Regression discontinuity in time (RDiT)

In our research context, a critical challenge is to identify the causal effects of recommendation algorithm upgrades and to find appropriate counterfactual estimates for those affected by these algorithm upgrades. Given that our focal algorithm upgrades are market-wide and that all complementors are subject to these quasi-natural experimental interventions, we lack an actual complementor control group to estimate counterfactuals. The RDiT design provides an identification framework for trustworthy inference in this case (Hausman & Rapson, 2018; Loh & Kretschmer, 2023).[6] We use the following panel regressions as our baseline models:

$$Y_{i,t} = \alpha_0 + \alpha_1 \text{PopRec}_t + \sum_{k=1}^{p} \omega_{1,k} t_{\text{A}}^k + \sum_{k=1}^{p} \omega_{2,k} \text{PopRec}_t \times t_{\text{A}}^k + \delta_i + \tau_t + \boldsymbol{Z}_{i,t}\boldsymbol{\gamma}_A + \varepsilon_{i,t} \quad (1)$$

$$Y_{j,t} = \beta_0 + \beta_1 \text{PersRec}_t + \sum_{k=1}^{p} \nu_{1,k} t_{\text{B}}^k + \sum_{k=1}^{p} \nu_{2,k} \text{PersRec}_t \times t_{\text{B}}^k + \delta_j + \tau_t + \boldsymbol{Z}_{j,t}\boldsymbol{\gamma}_B + \varepsilon_{j,t} \quad (2)$$

[6]RDiT is a special application of regression discontinuity design in which time serves as the running variable, and treatment begins at a particular threshold in time (Davis, 2008; Hausman & Rapson, 2018). The validity of RDiT relies on the smoothness assumption, which posits that the unobserved factors evolving over time change continuously around the time of the exogenous event (e.g., the algorithm upgrades in our context). This implies that no other concurrent events should cause a discontinuous jump in the outcome variables. In our research design, the pre-upgrade time trends capture the underlying evolution of the outcome, allowing us to project a counterfactual path into the post-event period. The treatment effect is thus identified by the discontinuity between the observed post-upgrade data and this extrapolated counterfactual. The quasi-natural experimental design based on RDiT is popular for examining the impacts of large-scale, exogenous events for its internal validity and ability to provide a clear, visual identification of causal impacts (Loh & Kretschmer, 2023; Ozturk et al., 2019), especially in recent digital platform research (Cao et al., 2024; Liu & Cong, 2023).

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TABLE 1 Descriptive statistics.

Variables	Panel A: Baseline → PopRec (195 days; 1,670,029 observations)				Panel B: PopRec → PersRec (184 days; 1,759,102 observations)			
	Mean	SD	Min.	Max.	Mean	SD	Min.	Max.
HHI of dish genres	6597.924	3224.410	362.426	10,000	6499.667	3299.381	362.426	10,000
HHI of adopted meat ingredients	4671.466	1940.841	0	10,000	4643.974	1962.825	0	10,000
HHI of adopted cooking methods	4470.286	2433.155	0	10,000	4445.827	2459.707	0	10,000
Number of new dishes	0.038	0.341	0	47	0.029	0.310	0	47
Number of new meat ingredients	0.014	0.167	0	20	0.011	0.156	0	20
Number of new cooking methods	0.018	0.203	0	19	0.014	0.181	0	31
Daily sales revenue (in CNY)	47.663	173.380	0	5865	42.278	172.120	0	11,435
Number of daily orders	1.119	4.068	0	244	0.943	3.811	0	187
Number of prior reviews	203.108	388.507	0	6524	218.143	436.245	0	7483
Average review stars (ratings)	4.364	1.455	0	5	4.367	1.449	0	5
Number of available dishes in the menu	32.088	37.714	1	780	32.973	39.512	1	887
Average dish prices (in CNY)	22.944	13.527	1	451	23.367	14.018	1	451
Value of coupon offered (in CNY)	0.846	3.284	0	144	0.735	3.165	0	140
Delivery radius (in meters)	2192.620	1299.370	500	10,000	2242.827	1329.400	500	10,000
Delivery fee (in CNY)	2.717	0.683	0	5	2.643	0.697	0	5
Number of kitchen tags	1.818	1.070	0	7	1.783	1.068	0	7
Complementor's gender (Male: 1, Female: 0)	0.278	0.448	0	1	0.282	0.450	0	1
Complementor's age level	41.548	11.838	20	60	41.193	11.910	20	60
Number of competitors per km^2 (density)	96.333	113.569	0	1124.271	103.661	123.805	0	1210.852
Proportion of buyers who disclosed gender	0.048	0.157	0	1	0.040	0.145	0	1
Proportion of buyers who disclosed age	0.091	0.242	0	1	0.075	0.223	0	1
Number of platform WeChat promo	0.736	0.821	0	3	0.634	0.753	0	3

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where the subscripts i and j denote each individual complementor in our dataset, and the subscript t denotes each day of the period during which the naïve geo-fencing interface, the PopRec algorithm, or the PersRec algorithm is implemented. $Y_{i,t}$ and $Y_{j,t}$ indicate the outcome of interest for complementors i and j on day t. As mentioned, in our models, the platform algorithm upgrades are exogenous to (i.e., unexpected by) both complementors and consumers, so the estimates are not endogenous, but rather show possible effects and allow for causal inference (Davis, 2008).

The running variables, t_A and t_B, are the numbers of days before or after the PopRec and PersRec algorithm upgrades, respectively. $\omega_{1,k}$, $\omega_{2,k}$, $\nu_{1,k}$, and $\nu_{2,k}$ capture flexible polynomial time trends with orders up to p. The interaction terms, that is, $\text{PopRec}_t \times t_A^k$ and $\text{PersRec}_t \times t_B^k$, allow for different polynomial fits on either side of the algorithm upgrade shocks (Cao et al., 2024). The optimal order p of the time polynomial is chosen according to the Bayesian information criterion (BIC) (Liu & Cong, 2023; Ozturk et al., 2019). δ_i and δ_j capture unobserved complementor fixed effects and τ_t captures the day-of-week effects to control for the calendar effects of weekend-weekday seasonality. To rule out alternative explanations, we use comprehensive sets of control variables, $\boldsymbol{Z}_{i,t}$ and $\boldsymbol{Z}_{j,t}$, whose descriptions are compiled in Table 1. $\varepsilon_{i,t}$ and $\varepsilon_{j,t}$ are the random error terms.

4.1.2 | Results for H1 and H2

We examine the impacts of the algorithm upgrades on complementor offering strategies to elucidate the orchestration function of recommendation algorithms. To provide a preliminary and intuitive demonstration of the algorithm effects, we first present the daily-level RDiT graphs in Figure 2. Specifically, we regress our dependent variables on all the control variables and seasonality factors, except for the time trends, and obtain the residuals. Using these residuals, we conduct nonparametric kernel regression estimations on both sides of the algorithm upgrades (Hausman & Rapson, 2018; Liu & Cong, 2023). Subsequently, we employ models (1) and (2) to parametrically estimate the effects of PopRec and PersRec on the offering strategy variables defined in Section 3, with the results presented in Table 2.

The discontinuities exhibited by the RDiT graph, along with the parameter estimation results, reveal nuanced trade-offs consistent with our hypotheses H1 and H2. Results in Table 2 suggest that implementing the PopRec upgrade yields a significant improvement in the HHI concentration of dish genres by 1.493 ($p = .000$), confirming its efficacy in promoting complementors' concentration and specialization of a few dish offerings, thus supporting H1(a). However, following the implementation of the PopRec algorithm, the daily introduction of new dishes by complementors decreases significantly ($p = .000$), supporting H1(b). This implies that although the PopRec algorithm encourages complementors to focus on a limited number of core products, such behavioral steering can come at the expense of the platform's generativity.

Exactly opposite dynamics emerge with the PersRec upgrade. As complementors are incentivized to introduce more new dishes ($p = .000$, supporting H2(a)), their HHI concentration of dish genres decreases by 0.011 ($p = .000$, supporting H2(b)). This offers evidence that they are pursuing greater new product introduction at the expense of concentration and specialization. In Supporting Information Appendices E1 and E2, we employ multiple alternative measures of the dependent variables and obtain consistent graphical and statistical evidence.

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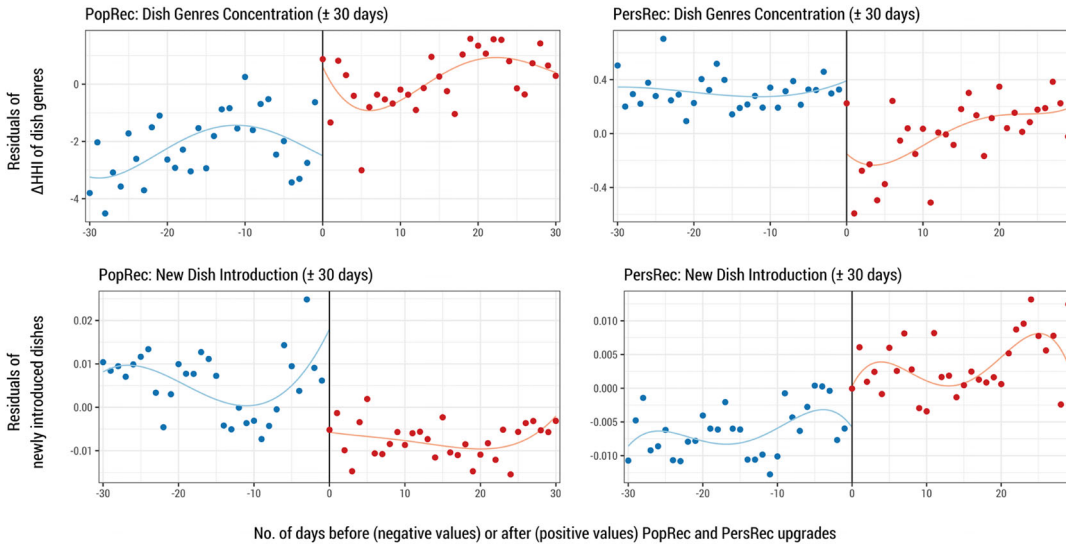

FIGURE 2 RDiT graphs for the strategic impacts. The RDiT graphs show the residuals of the dependent variables after removing the control variables and seasonality (i.e., day-of-week effects). The residuals are calculated using the entire data window, and then kernel regressions are estimated using only the residuals for days within a narrow bandwidth (i.e., an augmented specification) (Hausman & Rapson, 2018).

TABLE 2 Strategic impacts of PopRec and PersRec.

	(1)	(2)	(3)	(4)
Variable	**PopRec**	**PersRec**	**PopRec**	**PersRec**
DV	**ΔHHI of dish genres**	**ΔHHI of dish genres**	**New dishes**	**New dishes**
PopRec	1.493		−0.011	
	(0.264)		(0.002)	
PersRec		−0.894		0.006
		(0.156)		(0.002)
Constant	216.432	498.976	0.760	0.496
	(16.473)	(53.307)	(0.044)	(0.042)
BIC-selected order	1	1	2	2
Control variables	Included	Included	Included	Included
Day-of-week effects	Included	Included	Included	Included
Complementor fixed effects	Included	Included	Included	Included
Number of complementors	9587	10,462	9587	10,462
Number of observations	1,670,029	1,759,102	1,670,029	1,759,102
Within R^2	0.038	0.073	0.058	0.015

Note: Robust standard errors clustered at the complementor level are reported in parentheses.

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4.1.3 | Additional heterogeneity analyses

The substantial heterogeneity among the numerous complementors can influence both their motivation and capability to respond to platform design changes, thereby introducing additional orchestration complexity (Koo & Eesley, 2021; Loh & Kretschmer, 2023; Rietveld et al., 2021). We shed light on this by exploring how heterogeneity in complementors' prior status (more specifically, prior review rating and prior assortment size) and local competitive intensity[7] influences their strategic responses (see additional details regarding this variable in Supporting Information Appendix D), with the estimated results presented in Table 3.

Our results indicate that, for the PopRec upgrade, complementors with high prior ratings and facing intense competition are less responsive to the PopRec algorithm (i.e., exhibiting relatively lower offering concentration and a smaller reduction in new offering introduction). For the PersRec upgrade, complementors with a large assortment size are more responsive (i.e., exhibiting relatively higher offering diversification and more new offering introduction). Interestingly, complementors facing intense competition, while introducing a broader range of new offerings in response to PersRec, also tend to maintain specialization and concentration on a few core offerings. The theoretical expectations and discussion of results related to these analyses are compiled in Supporting Information Appendix C.

4.1.4 | Robustness checks

We conduct a variety of robustness tests to support our main findings. The potential issues and the corresponding proposed checks are summarized in Table 4 and detailed in Supporting Information Appendix E.

First, following Hausman and Rapson (2018), we examine the robustness of our RDiT specifications: (1) we validate the continuity of the covariates to alleviate concerns about precise platform manipulation (Liu & Cong, 2023); (2) we use various global polynomials to capture time trends; (3) we introduce lagged dependent variables as regressors to capture potential serial correlation; (4) we narrow the estimation window to enhance local randomization; (5) we complement the study with placebo tests, which do not reveal significant breakpoints before algorithm upgrades; and (6) we implement a "donut regression" to eliminate concerns about short-run selection, anticipation, and avoidance effects.

Second, we employ various methods to adjust the panel and sample settings: (1) we construct balanced panels for estimation by using only those complementors that were consistently active on the platform throughout the observation periods; (2) we include the *No Recommendation* phase as part of the pre-treatment condition for PersRec to estimate its effects; (3) we select a range of subsamples to mitigate the potential influence of outliers, ceiling effects, and the inclusion of professional complementors.

Finally, we adopt a series of alternative measures for complementors' offering strategies, which are then utilized to estimate our model. We also make various adjustments to our

[7]We measure the local competitive intensity by calculating the number of competitors per square kilometer within each complementor's service area (competitor density). Each complementors possess a fixed geographical location, with its service area centered on this location and delineated by a self-defined service radius. If the service areas of two complementors overlap, they are classified as competitors.

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TABLE 3 Heterogeneous strategic effects of the algorithm upgrades.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Variable	**PopRec**	**PopRec**	**PersRec**	**PersRec**	**PopRec**	**PopRec**	**PersRec**	**PersRec**
Moderator	**Prior review rating**		**Prior assortment size**		**Local competitive intensity**			
DV	**ΔHHI of dish genres**	**New dishes**	**ΔHHI of dish genres**	**New dishes**	**ΔHHI of dish genres**	**New dishes**	**ΔHHI of dish genres**	**New dishes**
PopRec	2.153	−0.013			2.335	−0.016		
	(0.317)	(0.002)			(0.347)	(0.002)		
PopRec × High Rating	−1.330	0.004						
	(0.377)	(0.002)						
PopRec × High Competition					−1.665	0.009		
					(0.385)	(0.002)		
PersRec			−0.570	0.003			−1.866	0.007
			(0.172)	(0.002)			(0.241)	(0.002)
PersRec × Large Assortment			−0.880	0.006				
			(0.231)	(0.002)				
PersRec × High Competition							1.966	−0.003
							(0.235)	(0.002)
Constant	216.601	0.760	499.000	0.497	215.440	0.766	501.932	0.493
	(16.492)	(0.044)	(53.319)	(0.042)	(16.468)	(0.044)	(53.472)	(0.042)
BIC-selected order	1	2	1	2	1	2	1	2
Control variables	Included	Included	Included	Included	Included	Included	Included	Included
Day-of-week effects	Included	Included	Included	Included	Included	Included	Included	Included
Complementor fixed effects	Included	Included	Included	Included	Included	Included	Included	Included

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TABLE 3 (Continued)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Variable	**PopRec**	**PopRec**	**PersRec**	**PersRec**	**PopRec**	**PopRec**	**PersRec**	**PersRec**
Moderator	**Prior review rating**		**Prior assortment size**		**Local competitive intensity**			
DV	**ΔHHI of dish genres**	**New dishes**	**ΔHHI of dish genres**	**New dishes**	**ΔHHI of dish genres**	**New dishes**	**ΔHHI of dish genres**	**New dishes**
Number of complementors	9587	9587	10,462	10,462	9587	9587	10,462	10,462
Number of observations	1,670,029	1,670,029	1,759,102	1,759,102	1,670,029	1,670,029	1,759,102	1,759,102
Within R^2	0.038	0.058	0.073	0.015	0.038	0.058	0.073	0.015

Note: Robust standard errors clustered at the complementor level are reported in parentheses. Complementors are classified according to the median of prior review ratings, prior assortment size, and local competitor density, yielding categories of high-rated/low-rated, large-assortment/small-assortment, and high-competition/low-competition.

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TABLE 4 Overview of the robustness checks.

Facets	Motivation/Concerns	Additional robustness checks
RDiT	To confirm the validity of the assumption that there is no precise platform manipulation of the assignment variable (time)	Covariate continuity test (RDiT plots on control variables)
	The time trends may be incorrectly specified	Regressions with polynomial global specifications
	There may be a potential serial correlation in the dependent variable (error term)	Regressions with lagged dependent variables
	It may be closer to local randomization at a narrower bandwidth setting	Samples from shorter time windows (±30 days, ±45 days, ±60 days)
	We may observe non-zero and statistically significant jumps at other discretionary cutoff points if RDiT does not work	Placebo test: false intervention time (−30 days, −60 days)
	To mitigate concerns about short-run selection, anticipation and avoidance effects	"Donut" regression dropping central samples (±1 day, ±5 days, ±10 days)
Variable construction	• The results may depend on specific data patterns and the construction of variables • The variables employed in the main analysis may not fully capture the strategic variations of complementors in offering strategies • A granular examination of the changes across multiple product dimensions	Alternative metrics for product concentration: • HHI of meat ingredients • HHI of cooking methods • HHI of customer-recommended dishes • Menu breadth of dish genres • Menu breadth of meat ingredients • Menu breadth of cooking methods Alternative metrics for new product introduction: • Newly developed meat ingredients • Newly developed cooking methods
Panel settings	The entry and exit of complementors in the market may disrupt the stability of the panel structure, thereby affecting the estimates	Sub-sample of complementors who were consistently active during the observation periods
	Showing effects in the entire observation period	Using the no-recommendation stage as the pre-treatment condition for the PersRec algorithm
Sample specification	The ceiling effect of 5-star complementors may affect the conclusion since they could not further improve their review ratings	Sub-sample excluding 5-star complementors
	Professional complementors may dominate market changes, thereby influencing the generalizability of conclusions	Sub-sample excluding "professional" complementors: • with a vast product assortment (>100, >75, >50 dishes) • with extreme transactions (maximum daily revenue larger than 5000, 3000, 1000 CNY)
Model and standard error specification	Unobservable heterogeneity may bias the estimates	Hierarchical Bayesian estimation

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TABLE 4 (Continued)

Facets	Motivation/Concerns	Additional robustness checks
	The new dishes variable: (1) is a non-negative integer; (2) exhibits right-skewness; and (3) is left-censored by zero. These issues may yield inconsistent estimates	Count models: (zero-inflated) Poisson and (zero-inflated) Negative Binomial
		Different transformation function specifications
		Tobit estimates
	Under certain conditions, the RE estimator can be more efficient than the FE estimator	Random effect estimation
	Standard error specifications: • Potential correlation within the same district • Potential cross-sectional correlation	Robust standard error clustered at district level
		Driscoll Kraay standard error

variable specifications, model specifications (e.g., random coefficient models, count models), and standard errors specifications. All of these tests yield consistent results for our main findings.

4.2 | Results for the impacts of AI recommendation algorithms on market equality

4.2.1 | Results for H3 and H4

To test H3 and H4, we operationalize market equality by assessing the performance parity among all complementors in the market. Specifically, we aggregate the total revenue and order amounts of each complementor before and after PopRec and PersRec upgrades. Subsequently, we estimate Lorenz curves for both revenue and order amounts, from which we calculate the corresponding Gini coefficients. Lorenz curves and Gini coefficients are principal metrics for characterizing income and wealth inequality (Gini, 1921; Lorenz, 1905), and are increasingly employed to describe sales distributions in online marketplaces (Brynjolfsson et al., 2010; Lee & Hosanagar, 2019). Figure 3 illustrates the changes in Lorenz curves and Gini coefficients observed over 30-day periods before and after the algorithm upgrades. To ensure robustness, we further report the results for 60-day periods in Supporting Information Appendix C.

Our results show that, after the implementation of PopRec, the revenue and order amounts across the market become more dispersed (indicated by a Lorenz curve closer to the 45-degree line of equality and a lower Gini index), suggesting a more equal revenue distribution on the platform (hence supporting H3). In contrast, following the implementation of PersRec, revenue and order amounts become more concentrated across the market on the platform (indicated by a Lorenz curve further from the 45-degree line of equality and a higher Gini index), thus exacerbating market inequality (hence supporting H4).

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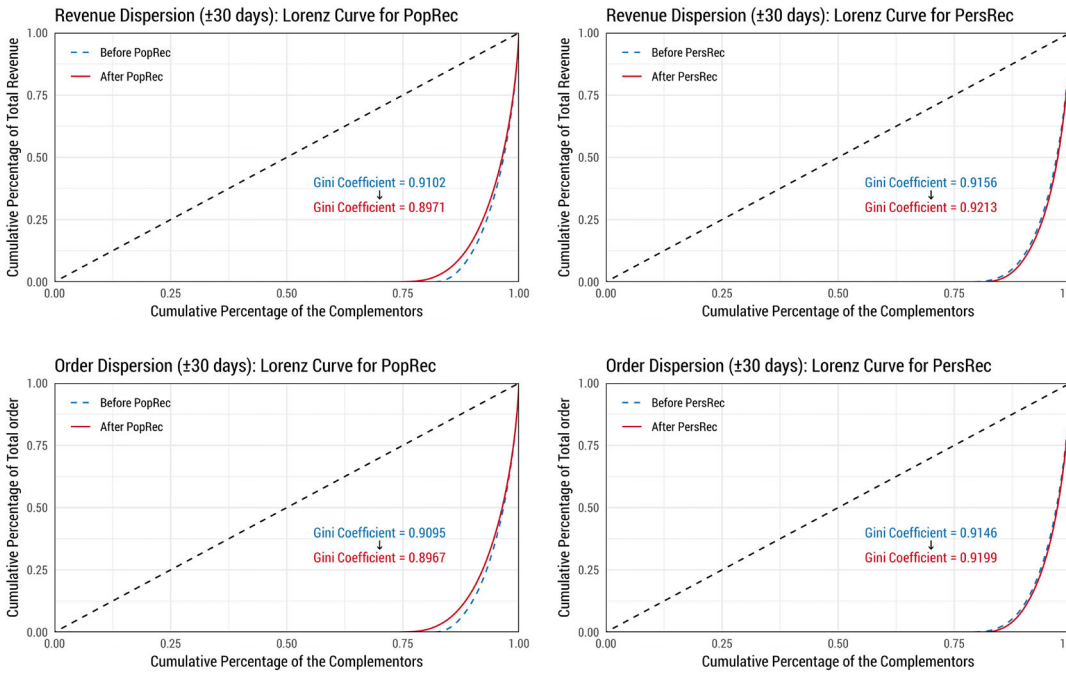

FIGURE 3 Lorenz curves for the revenue and order dispersion.

4.2.2 | Underlying dynamics

To further uncover the underlying dynamics behind these changes in revenue distribution, we compare the revenue changes of superstars with long-tail complementors before and after the algorithm upgrades. We use the quantiles of complementors' revenue prior to the algorithm upgrades to classify superstar and long-tail complementors and replace the dependent variables in Models (1) and (2) with the complementor's daily revenue.[8] The estimation results are presented in Table 5.

Columns (1)–(3) indicate that PopRec significantly reduces the revenue for prior superstars while boosting revenue for prior long-tail complementors, thereby leveling the competitive landscape. This finding is consistent with our expectation under Hypothesis H3, indicating that long-tail complementors, particularly those with high ratings but previously underexposed, experienced a significant revenue increase from PopRec (see results in Supporting Information Appendix C). In contrast, established superstars are more likely to be exposed to threatening shocks, resulting in significant revenue declines. Particularly, such threats may intensify their defensive and exploitative efforts to salvage past investments (Lavie et al., 2010), thereby diverting resources away from new product development. In other words, they may face more pronounced strategic trade-offs under the influence of PersRec. This dynamic is corroborated by the results in Supporting Information Appendix C, which reveal that superstars scaled back significantly more new product launches following the implementation of PopRec.

[8]The daily revenue variable is log-transformed to address right-skewness. Our results remain robust with the non-log-transformed dependent variables.

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TABLE 5 The micro-dynamics of revenue dispersion.

	(1)	(2)	(3)	(4)	(5)	(6)
Variable	**PopRec**	**PopRec**	**PopRec**	**PersRec**	**PersRec**	**PersRec**
DV	**Revenue**	**Revenue**	**Revenue**	**Revenue**	**Revenue**	**Revenue**
Sample	**Long-tail (≤80%)**	**Superstar (80%–90%)**	**Superstar (>90%)**	**Long-tail (≤80%)**	**Superstar (80%–90%)**	**Superstar (>90%)**
PopRec	0.007	−0.019	−0.075			
	(0.003)	(0.031)	(0.031)			
PersRec				0.007	0.136	0.234
				(0.002)	(0.025)	(0.034)
Constant	0.337	−0.341	0.359	0.392	0.042	0.400
	(0.048)	(0.218)	(0.299)	(0.077)	(0.196)	(0.393)
BIC-selected order	3	3	3	2	2	3
Control variables	Included	Included	Included	Included	Included	Included
Day-of-week effects	Included	Included	Included	Included	Included	Included
Complementor fixed effects	Included	Included	Included	Included	Included	Included
Number of complementors	7863	861	863	8537	962	963
Number of observations	1,359,528	150,623	159,878	1,432,448	157,978	168,676
Within R^2	.702	.711	.701	.743	.692	.624

Note: Robust standard errors clustered at the complementor level are reported in parentheses.

On the other hand, Columns (4)–(6) indicate that PersRec leads to significant revenue increases for both superstars and long-tail complementors. However, as PersRec disproportionately benefits superstars compared to the long-tail, it widens the revenue gap between these two groups. This finding aligns with our initial expectations when developing Hypothesis H4, suggesting that superstars are better equipped for rich, agile, and effective new product introduction, enabling them to adapt more rapidly to meet widespread, fragmented demand. The results in Supporting Information Appendix C further corroborate this, indicating that superstars engaged in substantially more new product development compared to long-tail complementors.

4.2.3 | Additional aggregate analyses

We further analyze the aggregate impacts of the two algorithms on the revenue generation across the entire platform (including both superstar and long-tail complementors), which bears significant implications for the platform owner's value capture and the stability of the platform

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marketplace. Our estimates indicate that PopRec leads to a significant decrease in the average revenue, indicating a substantial decline in the platform's overall economic performance. This implies that the boost PopRec provided to the long-tail complementors is insufficient to compensate for the revenue not realized by the superstars. In stark contrast, PersRec results in a significant increase in overall revenue on the platform. This disparity reveals a nuanced tension between platform equality and platform performance (see detailed analysis, results, and discussion in Supporting Information Appendix C).

5 | DISCUSSION

This study offers a novel perspective on how AI recommendation algorithms operate as platform orchestration mechanisms, shaping complementors' offering strategies and subsequent market equality. Leveraging two dominant AI recommendation algorithm upgrades on a leading food-sharing platform, we find that the PopRec algorithm incentivizes complementors to narrow their focus to a core set of offerings, but simultaneously impedes the introduction of new offerings. Moreover, it precipitates a decrease in the revenue for superstars while boosting the revenue for the long-tail, thereby fostering a more equal revenue distribution. In contrast, the PersRec algorithm stimulates complementors' engagement in new product introduction while reducing their product concentration. Although PersRec facilitates a widespread increase in revenues for both superstar and long-tail complementors, the superstars disproportionately benefit, which exacerbates the inequality in market revenue distribution.

5.1 | Theoretical contributions

A key contribution of this study is to highlight the value of recommendation algorithms in orchestrating the tradeoff between platform generativity and control. Although a preponderance of literature has highlighted the role of platform owners in guiding platform complementors toward ecosystem objectives (Jacobides et al., 2018; Van Angeren & Karunakaran, 2023; Zhang et al., 2022), it remains largely unexplored how platform owners can design and leverage orchestration mechanisms to navigate the generativity-control balance and how the persistent tension between the two can inhibit platforms' orchestration practices (Engert et al., 2025; Staub et al., 2022; Yoo et al., 2012). We articulate a novel pathway whereby platforms, by strategically deploying and upgrading recommendation algorithms, can reconfigure the incentive landscape for complementors, thereby nudging their offering strategies into alignment with the platform's generativity-control objectives, without impinging upon their autonomy or contravening community norms (Hunt et al., 2025; Tiwana, 2013; Tiwana et al., 2010). Relatedly, we also reveal how this strategic alignment is constrained by complementors' strategic tradeoffs, and consequently, by the inherent tension between platform generativity and control.[9] In this sense, we respond to Rietveld and Schilling's (2021) call to investigate the role of orchestration mechanisms in balancing seemingly conflicting tensions on platforms.

[9]We draw on a strategic offerings perspective to illuminate how generativity and control pull platform owners in opposing directions, while acknowledging that generativity extends beyond mere the introduction of new products and that control encompasses more than just specialization and efficiency improvements from offering concentration.

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Second, this study bridges the discourse on emerging AI technologies with the literature on digital platform ecosystems. Despite the growing prevalence of AI-based applications on digital platforms, scholarly understanding remains limited on the mechanisms through which AI fosters new forms of digital ecosystem management and empowers distributed ecosystem actors (Hunt et al., 2025; Mayer et al., 2025). We delineate the managerial functions manifested by AI recommendation algorithms, which operate through data-driven algorithmic decision-making to automate and tailor selective promotion, thereby demonstrating the capabilities and value of predictive AI in ecosystem orchestration. Moreover, adopting a platform-ecosystem perspective enables us to interpret AI's strategic potential from a broader vantage point. The dominant AI narrative in strategic management focuses on how AI, as *internal* resources or capabilities, can be leveraged by organizations or individuals to achieve strategic goals and sustain a competitive edge ("strategy *with* AI") (Berg et al., 2023; Choudhury et al., 2020; Doshi et al., 2025; Jia et al., 2024; Krakowski et al., 2023; Raisch & Fomina, 2025). An underrated perspective is how AI, as an *external* enabler, guides and shapes strategic decision-making and alters the competitive landscape ("strategy *under* AI"), which is crucial for revealing the market orchestration potential and concomitant risks of AI applications (Grote et al., 2024). We contribute to this external perspective by demonstrating that competitive platform markets will spontaneously undergo adaptive evolution driven by the incentives embedded in AI algorithms, thereby triggering shifts in market value structures and a delicate trade-off between platform performance and equality.

Third, by demonstrating the downstream impacts on market equality, we extend our insights into the complex value dynamics that emerge in platform markets when AI acts as market orchestrators (Koo & Eesley, 2021; Sun et al., 2024; Zhang et al., 2021). We show that the skewed impacts of algorithms on complementors reshape the competitive landscape and alter the distribution of gains within the marketplace, notably revealing some thought-provoking equality consequences. While PopRec operates through relatively centralized promotion, it primarily benefits more long-tail complementors, fostering a more equal distribution of market revenue. This, however, comes at the cost of a reduction in the platform's aggregate revenue. Conversely, PersRec, though ostensibly enhancing broad visibility and boosting aggregate revenue, simultaneously exacerbates revenue disparity due to the advantages of superstars in adaptability and strategic adjustment. A key implication is that platforms may sacrifice some performance when using PopRec to promote specialization and quality control. Conversely, when employing PersRec to foster generativity, high vigilance is required concerning the potential for worsening market inequality. These findings also indicate that the market dynamics and welfare effects stemming from AI algorithms can be more intricate and may compel platforms to navigate a delicate trade-off between economic performance and market equality. These insights add to prior research that AI-induced alterations in complementor dynamics will elicit the co-evolution of platforms' managerial orientations, thereby influencing the evolutionary dynamics of the platform ecosystem (Rietveld et al., 2020; Tiwana et al., 2010). In this manner, this study also contributes to the burgeoning literature on AI-centric marketplace design and AI-catalyzed market evolution (Furman & Seamans, 2019; Hillebrand et al., 2025; Jacobides et al., 2021).

5.2 | Managerial implications

This study also draws relevant managerial implications for practitioners. The strategic engineering of AI recommendation algorithms can serve as a low-cost yet effective platform

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orchestration mechanism. This algorithmic orchestration underscores the significant potential of AI technology in managing platform ecosystems. AI technology enables platform owners to promote specific complementors in an automatic, data-driven, and scalable manner. Automation facilitates the dynamic and rapid capture of changes among complementors; the data-driven nature facilitates precision orientation, thereby mitigating the subjectivity and bias of direct human intervention; and scalability allows the system to adapt to large-scale, high-frequency fluctuations in users and complements on the platform, empowering platform owners to design and calibrate algorithmic incentives at scale. Particularly, recommender systems, acting as the conduit for AI orchestration, allow promotions based on algorithmic decision-making to directly reach individual consumers. These characteristics render algorithmic promotions highly timely, objective, reliable, and accurate, thereby providing immediate, potent, and extensive strategic incentives for complementors.

Essentially, AI algorithmic orchestration relies on the self-regulation of complementors without involving formal structures, thereby circumventing direct resistance from complementors and associated regulatory risks. However, our findings also reveal that this seemingly "light-touch" orchestration approach can precipitate additional challenges, such as the PopRec-induced diversity diminution and the PersRec-induced proliferation of low-quality product introduction. This implies that, in extreme cases, algorithmic orchestration with a myopic and singular focus might work out as a case of being "penny-wise and pound-foolish." Platform owners should meticulously monitor the algorithmic outcomes and identify which complementors bear the brunt or experience the most significant trade-offs. Armed with this assessment, they should then offer adaptive interventions (e.g., alerts, visibility quotas, or training) to notify or support specific complementors. Our analysis indicates that AI recommendation algorithms predominantly promote particular offering strategies, rather than empowering complementors to independently overcome the inherent strategic trade-off. Additional algorithmic calibrations and interventions aimed at promoting ambidexterity could be beneficial. Extant literature proposes several pathways to ambidexterity, such as temporal separation and domain separation (Lavie et al., 2010; Lavie & Rosenkopf, 2006; Stettner & Lavie, 2014). Platforms could dynamically, sequentially, and alternately modulate the algorithmic preference of recommender systems within different timeframes, thereby managing the temporal shifts between complementors' offering activities (a *time*-based approach). Alternatively, platforms could directly deliver targeted messages to complementors or subtly modulate algorithmic mechanisms to guide their strategic focus across different domains (e.g., target markets, raw material diversity, taste diversity), thereby achieving strategic balance through cross-domain complementarity (a *domain*-based approach) (Lavie & Rosenkopf, 2006; Papachroni & Heracleous, 2020).

Furthermore, platform owners should be cognizant of the ethical and evolutionary risks potentially engendered by AI algorithms. The bilateral, or even multilateral, evolutionary complexity triggered by AI orchestration could reshape the platform market landscape in a profound and unforeseen manner. Our study provides a relevant observation: due to disparities in adaptive capacity, widely deployed personalized recommendations can exacerbate economic inequality among platform complementors. This aligns with the findings of recent studies on recommendation algorithms (Lee & Hosanagar, 2019; Tan et al., 2017). Such inequality can not only trigger ethical controversies but could also, over the longer term, lead to the gradual crowding out of some long-tail complementors, thereby altering the platform's value structure (Rietveld et al., 2019; Rietveld & Schilling, 2021; Wareham et al., 2014). As the nexus between various AI-driven personalized applications and digital platforms becomes more tightly

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coupled, the winner-takes-all effects and competitive isomorphism on the supply side warrant closer scrutiny. In this sense, our findings also underscore the necessity for regulatory authorities to consider AI's far-reaching ramifications in policy formulation to ensure that technological advancements align with commercial objectives and societal needs (Grote et al., 2024; Sun et al., 2024).

5.3 | Limitations and future research

This study has several limitations that can pave the way for further research. One limitation pertains to the scope of algorithms examined. While algorithms in our study represent two prevalent recommendation paradigms, other algorithmic variants and other potential processes of algorithmic evolution warrant further investigation. An interesting direction arises from tightening data-regulation policies: as shifts from personalized to non-personalized modes become more common (Sun et al., 2024), it remains to be explored whether complementors will mount strategic responses symmetric to those we identify. Moreover, with the rapid advancement of AI technologies (e.g., generative AI), it remains an open question whether more AI applications on digital platforms could demonstrate orchestration capabilities. Another limitation is that the specificity of our research context may restrict the generalizability of our conclusions. Supporting Information Appendix B provides a preliminary overview of how industry- and platform-specific features could affect our conclusions. In addition, while our algorithms are predicated on a complementor-based system, many digital platforms utilize product-based recommender systems (see a comparison of these two system types in Supporting Information Appendix B). A promising direction for future research lies in comparing the orchestration capabilities of algorithms across different industries, platforms, and system designs.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

We acquire a proprietary dataset from a major food-sharing digital platform. The data provider embargo data sharing due to commercial restrictions. We may provide synthesized data to replicate all results upon request.

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